

HW Testing

Electronics Project PM4: CW Radar

Abstract

Testing, and especially documenting tests, is often undervalued. Even if a prototype works in the laboratory this does not guarantee that it works in the field. Designing reproducible test procedures with a good coverage helps to reduce the costs of warranty and increase the profit margin. This “checklist” tells how to prepare, run and document tests on an electronic circuit.

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1. Purpose of Testing

- Verify or proof that the requirements and expectations on the Device Under Test (DUT) are reliably met
- Reduce the risk of failures
- Reduce the costs of warranty and increase the profit margin
- For a prototype: Make sure that the DUT is ready to be integrated into a bigger system

2. Design for Test

Testing starts with the hardware design!

- The tests should reflect the requirements and specifications. Ideally, the tests include the ones done while commissioning at the customer.
- When designing a circuit, identify its critical aspects and find a way to test them.
- Where meaningful a circuit can be extended for better testability:
 - Add test pins, test pads, jumpers, 0Ω resistors.
 - Connect Enable/Disable pins of ICs via pull-up or pull-down resistors to power or ground to be able to deactivate an IC for test purposes.
 - Add a jumper in the supply conductor of a sub-circuit. Thus, it is possible to test every sub-circuit independently.
 - However, ensure that such built-in test circuitry has no negative influence on the main functionality of the circuit (in particular when fast signals and/or higher frequencies are involved ...)
- Design the PCB to support testability:
 - Place the relevant nodes accessibly for oscilloscope probes, ...
 - Include GND pins for the oscilloscope and function generator.
 - Add some supporting text on the PCB such as signal labels, jumper settings, ...
 - Here again, make sure that these additions have no negative impact on the main function of the circuit. Avoid long lines, don't lose compactness ...

3. Test Plan

A good test plan covers all essential tests and minimizes redundant tests.

The test plan should include all information required to replicate the covered tests at a later time and get consistent, comparable test results.

It's good practice to build the test plan as a list of test cases. For each item in the list:

- Define its purpose. I.e. name the related requirement or the specific functionality to be measured and verified
- Specify the required test equipment and test circuits
- List the test conditions, such as supply voltage, input signals, output load, temperature, any settings in the DUT (jumpers, IC registers, ...) or in the test equipment
- Determine the expected results
- Define the conditions to pass the test

Check the test plan for:

- Consistent and well balanced test depth: The intensity of testing should directly align with the criticality of the functionality being verified.
- Useful test sequence: The sequence of tests performed should be feasible and reasonable. See also next section 4 Safety First.

The test plan must be finished before the tests are started and not afterwards! However, it is common practice, to modify a test plan during the tests.

Often it is possible to use a table with about five columns with the above aspects and a row for each test case. Below is an example for a signal amplifier:

Test item, Description	Test conditions	Expected, Tolerance	Measured result	Comparison, verdict	Remarks, Discussion
DC level of output	No input signal VCC = 3.3V	Output signal: 1.250V ±5mV	1.252V	deviation +2mV passed	
Gain	Input signal: 1mVpp 1kHz sine	Output signal: 250mVpp Gain: 240..270	234mVpp Gain = 234	Circuit works, but out of tolerance failed	Check resistor tol. Continue with tests
Linearity and noise	Input signal: 1mVpp 1kHz sine	THND < 2%	THND = 3%	failed	Cause unknown: Check PCB layout, Capacitor type: NP0? Abort test
Stability with capacitive load	Input signal: 1mVpp 1kHz rect 100pF on output	Output signal overshoot < 10%			

As can be seen, this table is also helpful for documenting the test procedure.

4. Safety First

Always follow the principle: Safety First!

Personal safety: Prevent risk of electric shock or any other danger due to high voltages or high currents, heat, intensive light, sound, ...

Prevent risk of damage of the DUT or the test equipment.

Before switching on the power supply do a visual inspection of the PCB (i.e. the DUT):

- Soldering, no short circuits, correct orientation of the components (electrolytic capacitors can explode with wrong polarisation), no missing components, correct values ...
- Verify the test setup: Correct power supply (voltage, current limits, polarity), instrument connection and configuration, ...

Start with safe measurements.

- Start with voltage, current and power limited to the lowest possible values.
- If possible, divide the whole circuit into sections (sub-circuits), operate and test them separately before doing the overall test. Start with the low power sections.

5. Run the Tests

Perform the tests as outlined in the test plan. Throughout the testing process, it should therefore always be clear what outcomes are expected.

Write a test protocol while doing the tests. It should include:

- All information about the conditions and results of the measurement. E.g. tables with raw data, generated log files or graphs
- Final verdict: test case is either passed or failed
- Comments on any noteworthy observations: unexpected results, special insights, instabilities ...
- Formal information: who did the test, date, time, location, identification of equipment used

General rule: Using the test plan and the test protocol, it should be possible to repeat a measurement and reproduce the results.

When a test has failed, decide whether the test session can be continued or whether it has to be aborted.

After correcting an error, at least all the tests depending on the affected component, must be repeated and the results must be documented again.

6. Test Analysis

Test analysis after the test session may include:

- Post processing of the test results
- Presentation of the results e.g. as graphs, tables, ...
- Detailed comparison against the expectations, simulations, datasheets, ...
- Finding reasons for deviations and errors
- Commenting and discussing the results
- Make a final verdict: passed or failed